

The University of Zambia  
Department of Mathematics and Statistics  
Mat 1110 - Foundation Mathematics and Statistics for  
Social Sciences

**Tutorial Sheet 5 - Polynomial & Rational Functions**

**April, 2023.**

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1. Use **long division** to divide each of the following polynomials by the given divisor.

|                                      |                |
|--------------------------------------|----------------|
| (a) $f(x) = -x^2 + 3x + 5,$          | $x^2 + 1$      |
| (b) $f(x) = 5x^3 - 19x^2 + 16x - 4,$ | $x - 5$        |
| (c) $f(x) = 2x^3 + 6x^2 - x + 3,$    | $2x^2 - 2$     |
| (d) $f(x) = -2x^4 + 3x^2 + 1,$       | $x^2 - 2x + 3$ |

2. Use **synthetic division** to divide each of the following polynomials by the given divisor, and write the polynomial in the form

$$p(x) = q(x)(x - a) + r,$$

where  $q(x)$  is the quotient and  $r$  is the remainder.

|                                      |           |
|--------------------------------------|-----------|
| (a) $p(x) = x^3 - 7x^2 - 4x + 28,$   | $x - 3$   |
| (b) $p(x) = 5x^3 - 6x^2 - 28x - 2,$  | $5x + 10$ |
| (c) $p(x) = x^3 - 10x^2 - 31x - 30,$ | $x - 11$  |
| (d) $p(x) = 6x^3 + x^2 - 21x - 10,$  | $2x + 1$  |

3. Factorize completely each of the following polynomials:

|                                   |                                            |
|-----------------------------------|--------------------------------------------|
| (a) $f(x) = 4x^3 - 8x^2 - x + 2$  | (d) $f(x) = x^3 - 1$                       |
| (b) $f(x) = 2x^3 + 3x^2 - 3x - 2$ | (e) $f(x) = x^3 + 1$                       |
| (c) $f(x) = x^3 - 2x^2 + 2x - 4$  | (f) $f(x) = x^4 - 6x^3 - 11x^2 + 24x - 28$ |

4. Solve each of the following equations:

|                               |                                     |
|-------------------------------|-------------------------------------|
| (a) $x^3 + x^2 - x - 1 = 0$   | (d) $x^2(2x + 1) = 13x - 6$         |
| (b) $2x^3 + 3x^2 - 3x = 2$    | (e) $x^4 - 6x^3 = 11x^2 - 24x - 28$ |
| (c) $x^3 + 6x^2 + 5x - 2 = 0$ | (f) $x^3 - 8 = 0$                   |

5. Let  $f(x) = 4x^3 - 7x - 3$ ,
- Find the remainder when  $f(x)$  is divided by  $(x - 2)$ .
  - Show that  $(2x + 1)$  is a factor of  $f(x)$ .
  - Hence, or otherwise, solve the equation  $4x^3 - 7x - 3 = 0$ .
6. (a) Given that  $x - 1$  and  $x + 1$  are factors of  $f(x) = ax^3 + bx^2 - 3x - 7$ , find the value of  $a$  and of  $b$ .
- (b) Let  $f(x) = x^3 + ax^2 + bx + 6$ . Given that the remainders when  $f(x)$  is divided by  $x + 1$  and  $x - 2$  are 20 and 8 respectively, find the value of  $a$  and of  $b$ .
7. Sketch each of the following functions, clearly indicating the  $x$  and  $y$  intercepts.
- $f(x) = x^3 - 13x + 12$
  - $f(x) = 15 + 5x - 3x^2 - x^3$
  - $f(x) = x^3 - 4x^2 + x + 6$
  - $f(x) = 4x^3 - 12x^2 + 5x + 6$
  - $f(x) = x^3 - 6x - 5$
  - $f(x) = -x^3 + 10$
8. For each of the following rational functions, find/state the domain and the range. Hence sketch the graph.
- $f(x) = \frac{2-x}{x+4}$
  - $f(x) = \frac{3x+6}{x-1}$
  - $f(x) = \frac{x+1}{x-1}$
  - $f(x) = \frac{2x+5}{x-1}$
9. Find the partial-fraction decomposition of the following rational functions:
- $\frac{1}{(x+5)(x+4)}$
  - $\frac{x+7}{x^2+3x+2}$
  - $\frac{2x^2-9x-35}{x^3+2x^2-5x-6}$
  - $\frac{x^2-1}{x(x^2+1)}$
  - $\frac{x-3}{x^3+3x}$
  - $\frac{2x}{(x+1)^2(x^2+1)}$
  - $\frac{x^2+1}{x^2-1}$
  - $\frac{4x^3+10x+4}{x(2x+1)}$
  - $\frac{2x^4+3x^2+1}{x^2+3x+2}$
  - $\frac{x^2+2}{(x+2)^2(x+3)}$
  - $\frac{4x}{x^3+4x^2+5x+2}$